

Curriculum Learning With Counterfactual Group Relative Policy Advantage For Multi-Agent Reinforcement Learning

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Abstract—Multi-agent reinforcement learning (MARL) has achieved strong performance in cooperative adversarial tasks. However, most existing methods typically train agents against fixed opponent strategies and rely on such meta-static difficulty conditions, which limits their adaptability to changing environments and often leads to suboptimal policies. Inspired by the success of curriculum learning (CL) in supervised tasks, we propose a dynamic CL framework for MARL that employs an self-adaptive difficulty adjustment mechanism. This mechanism continuously modulates opponent strength based on real-time agent training performance, allowing agents to progressively learn from easier to more challenging scenarios. However, the dynamic nature of CL introduces instability due to nonstationary environments and sparse global rewards. To address this challenge, we develop a *Counterfactual Group Relative Policy Advantage* (CGRPA), which is tightly coupled with the curriculum by providing intrinsic credit signals that reflect each agent’s impact under evolving task demands. CGRPA constructs a counterfactual advantage function that isolates individual contributions within group behavior, facilitating more reliable policy updates throughout the curriculum. CGRPA evaluates each agent’s contribution through constructing counterfactual action advantage function, providing intrinsic rewards that enhance credit assignment and stabilize learning under non-stationary conditions. Extensive experiments demonstrate that our method improves both training stability and final performance, achieving competitive results against state-of-the-art methods. The code is available at <https://github.com/NICE-HKU/CL2MARL-SMAC>.

Index Terms—Multi-agent reinforcement learning, curriculum learning, learning non-stationary, group relative policy optimization, counterfactual policy advantage, intrinsic reward shaping.

I. INTRODUCTION

IN real-world artificial intelligence (AI) applications, complex problems can be formulated as multi-agent cooperative decision-making tasks, where multiple intelligent agents collaborate to achieve shared objectives [1], [2]. Representative examples include traffic signal control, autonomous driving systems, communication resource allocation, and military simulation scenarios [3]–[6]. Multi-agent reinforcement learning

(MARL) has emerged as an effective framework for addressing these problems by enabling agents to learn coordinated strategies through interaction [7]. Notably, recent advancements in MARL have led to groundbreaking achievements of multi-agent cooperative adversarial research, with a variety of novel algorithms achieving state-of-the-art (SOTA) results in benchmark environments such as StarCraft II micromangement¹.

Previous studies have continuously addressed several existing key challenges in MARL, such as sparse global rewards, inefficient credit assignment among agents and insufficient environmental exploration [8]–[11], leading to substantial progress in algorithm design and performance. However, beyond these well-studied issues, a more fundamental and often overlooked limitation remains: most MARL models are trained under meta-stationary environmental conditions, which restrict policy adaptability. As highlighted in recent reviews [12], [13], widely adopted MARL benchmarks often assume fixed opponent behavior². For instance, in the challenging StarCraft-II multi-agent environment of the *StarCraft multi-agent challenge* (SMAC)³ benchmark, the game engine configures scripted opponents with predefined difficulty levels. Lower levels (e.g., Levels 1–3) serve as the ‘Easy’ modes, while higher levels (e.g., Levels 8–10) employ ‘Cheating’ modes with access to global vision, faster reactions, and additional resources. However, the constructed SMAC environment [14], [15] through its game engine API, exclusively defaults to Level 7 difficulty across all SMAC map scenarios [8]–[11].

While such meta-stationarity environments help maintain training stability and consistent policy convergence during training, their reliance on fixed opponent capabilities (e.g., the default Level-7 difficulty) poses a critical limitation: agents often overfit to static conditions and fail to generalize across dynamic scenarios. Consequently, agents frequently converge to suboptimal local solutions that perform well under static conditions but fail to adapt beyond them. Unlike the single-agent setting, MARL introduces additional challenges including *environmental non-stationarity*, *exponential joint action space growth*, *credit assignment rationality*, and *partial ob-*

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¹PySC2 - StarCraft II Learning Environment: <https://github.com/google-deepmind/pysc2>

²Here, we introduce a novel technical concept for MARL learning, namely “*environmental meta-stationarity of MARL*”, referring to the static, unchanging overall difficulty level of a task. Distinct from the inherent dynamicity of environmental states that change in response to agents’ actions during training, meta-stationarity describes the constant global challenge level maintained throughout all training episodes. For instance, in military simulation scenarios, the opponent’s AI capability remains fixed. Similarly, in traffic control tasks, the overall traffic flow complexity stays constant across episodes.

³SMAC and SMACv2 GitHub: <https://github.com/oxwhirl/smac>, and <https://github.com/oxwhirl/smacv2>.

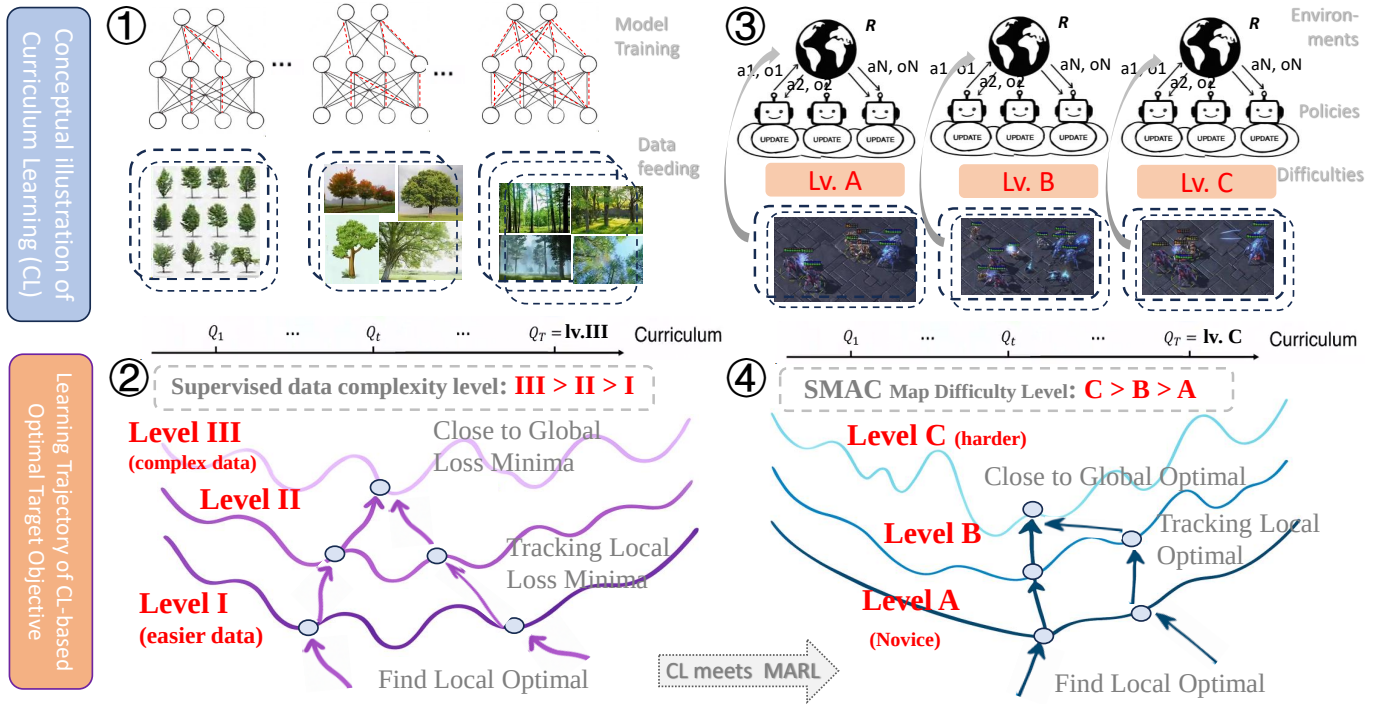


Fig. 1: *When CL Meets MARL*: A comparative illustration of training paradigms between CL-based supervised learning and CL-based MARL. Part 1 and Part 2 represent the traditional CL framework for supervised learning (e.g., CV, NLP) with labeled data and its optimization trajectories (supervision loss minimization), and Part 3 and Part 4 are the CL-based MARL training framework in the SMAC adversarial environment (2s_vs_3z map) and corresponding optimization trajectories (optimal policy exploration).

servability. These complexities are further exacerbated under fixed environmental dynamics. This observation motivates us to consider a fundamental question:

“Can we design a human-inspired progressive learning paradigm for the cooperative adversarial tasks of MARL that mimics the developmental stages of human skill acquisition to overcome the ‘environmental meta-stationarity’ limitation?”

In recent years, curriculum learning (CL) has gained significant attention as a progressive learning paradigm that mimics human learning process by moving from easier to more difficult tasks, particularly succeeds in supervised learning, e.g., computer vision (CV) and natural language processing (NLP) [16]. As demonstrated in the Part 1 of Fig. 1, compared with direct training on the entire dataset, CL strategically partitions the training data into difficulty-graded subsets, progressively increasing challenge levels during training, mirroring human educational curricula. From an optimization perspective shown in the Part 2 of Fig. 1, the training process first converges to a local minimum on a simpler and easier subset, and as the curriculum progresses, it continues to follow these minima while adapting to more complex objectives. This mechanism ensures sustained generalization improvement, ultimately enabling the model to approximate the global optimum.

Inspired by these advantages, we systematically investigate CL’s potential for optimizing multi-agent cooperative adversarial MARL training. To the best of our knowledge, no research has been conducted on implementing CL for MARL’s cooperative adversarial tasks such as SMAC before this work. Our

research effectively mitigates the long-overlooked limitations caused by the ‘environmental meta-stationarity of MARL’ phenomenon, poor generalization and local optima convergence pitfalls. Specifically, when applying CL to MARL, we identify key parallels with traditional supervised CL approaches. As illustrated in the Part 3 of Fig. 1, progressively increasing environmental difficulty for multi-agent systems enables policy learning to follow the same easy-to-hard progression. From a MARL policy optimization perspective (Part 4 of Fig. 1), agents first converge to local optima in simpler and easier environments. As the task difficulty escalates, the policies continuously adapt by tracking solution manifolds across progressively challenging environments, ultimately approaching the global optimum in the target difficulty environment.

To make CL effective in the MARL context, two fundamental components must be addressed, i.e., the design of a *difficulty measurer* and a *training scheduler* [16]. The fundamental challenges in CL lie in the implementation of the difficulty Measurer and the determinization of the training scheduler [16]⁴. In CL-based MARL, the difficulty measurer defines how to modify the task complexity, which can be achieved by tuning environment or task difficulty through hyperparameter configuration (e.g., varying opponent AI levels in SMAC’s StarCraft II scripting interface). The MARL

⁴In CL-based supervised learning (e.g., CV and NLP). The difficulty measurer focuses on partitioning the dataset into different difficulty levels and classifying each supervised sample. And the training scheduler determines when and how to introduce harder samples.

training scheduler determines when to adjust overall difficulty, where evident signals like overall rewards and task completion rates (e.g., SMAC rewards and win rates) serve as criteria for comprehensive task difficulty adjustment.

To implement this, we design a statistical-based adaptive training scheduler, namely ‘*flexible task difficulty scheduler*’ (**FlexDiff**), that dynamically modulates MARL task difficulty to optimize policy learning. Specifically, **FlexDiff** dynamically adjusts environmental task difficulty based on real-time agent performance metrics, particularly the rewards and task completion rates. The proposed module integrates three components, i.e., (1) momentum-driven adjustment, capturing persistent performance trends, (2) sliding window analysis, evaluating short-term stability via mean and std metrics, and (3) dual-curve monitoring, simultaneously tracking reward progression slopes and task completion rate derivatives. By synthesizing these multi-scale features through adaptive thresholding, the scheduler self-regulates difficulty levels in real time, achieving flexible curriculum adaptation to emergent agent capabilities while maintaining training stability against local optima⁵.

While CL improves MARL agents’ generalization, merely increasing difficulty via training feedback is suboptimal due to (1) post-transition reward sparsity and (2) policy rigidity. Agents converging to Difficulty I often fail to adapt to Difficulty II. We address this by prioritizing high-contributing agents via optimized credit assignment. Specifically, inspired by *group relative policy optimization* (GRPO) [17] and *counterfactual multi-agent policy gradients* (COMA) [18] algorithms, we propose a novel *Counterfactual Group Relative Policy Advantage* (**CGRPA**) algorithm that evaluates intra-agent action advantages by quantifying each agent’s environmental contribution through comparison between actually executed actions from empirical trajectories, and counterfactual actions that were possible but not actually selected. This algorithm helps identify key contributors at each timestep, thereby stabilizing convergence during difficulty transitions. **CGRPA** uniquely combines GRPO’s group optimization algorithm with COMA’s counterfactual theory, effectively addressing the non-convergence issues and significantly enhancing the agents’ policy optimization speeds. To our knowledge, this is the first technical integration of GRPO with COMA’s counterfactuals in MARL. Through extensive experiments on SMAC, we demonstrate that **CGRPA** effectively mitigates policy divergence risks during environmental difficulty transitions and enhances the coordinated implementation of CL in MARL.

In summary, our work builds directly off of the idea of CL paradigm and reward credit assignment enhancements. Through rational technological innovations, we have effectively

overcome the widespread challenge of environmental meta-stationarity in MARL. The primary contributions of this paper are summarized as follows:

- Through systematic analysis of existing MARL research, we firstly identify the prevalent challenge of ‘environmental meta-stationarity’ in MARL, which fundamentally limits agents’ policy generalization and constrains agents’ ability to discover globally optimal joint policies.
- To address this issue, inspired by CL from supervised learning, we propose a novel adaptive *training scheduler* for MARL, namely **FlexDiff**, that dynamically adjusts environmental task difficulty during training. To the best of our knowledge, this represents the first integration of CL training paradigm into the multi-agent cooperative adversarial scenarios of MARL, particularly in the testbed of StarCraft-II unit micromanagement, SMAC.
- To mitigate the non-convergence risks of MARL introduced by CL integration, we propose the **CGRPA** algorithm, a novel approach that combines GRPO’s group optimization with COMA’s counterfactual reasoning. By evaluating intra-agent advantages through the comparison between empirical and counterfactual actions, **CGRPA** stabilizes policy adaptation during difficulty transitions.
- Through extensive empirical evaluation across multiple SOTA MARL baselines, such as QMIX [19], OW-QMIX [8], DER [20], and EMC [9], our CL-enhanced MARL approach demonstrates significant performance improvements, achieving competitive results compared to SOTA algorithms. For instance, experiments on SMAC benchmark show that QMIX’s win rate remarkably surges from 45-60% to 75-90% in 5m_vs_6m map. These results not only validate the effectiveness of CL-MARL but also reveal CL’s significant potential for MARL research.

II. RELATED WORK

MARL, as an extension of RL under a single-agent context to more complex cooperative scenarios, has consistently garnered significant attention. Compared to single-agent RL, MARL inevitably faces several issues, such as: *Non-stationarity in Global States* [21]–[25], *Credit Assignment Problems* [8], [18], [19], [26], [27], *Partial Observability* [28]–[31], and *High-Dimensional Action Space* [32]–[35]. Additionally, other critical challenges in MARL, such as *Exploration-Exploitation Trade-off* [9], [36], [37], and *Scalability and Generalization* [38], have been continuously researched as they play an essential role in the effective MARL optimization. Next, we will provide a brief overview of recent advancements in addressing these challenges.

For dealing with non-stationarity of global states, Nekoei et al. [24] proposed a multi-timescale learning approach in decentralized cooperative MARL, where agents update their policies at different rates, improving convergence speed and minimizing non-stationarity. Wei et al. [23] formalized the non-stationary environment using pessimistic transitions and introduced prudent Q-learning (PQL) to control multi-agent systems independently in such environments. Sun et al. [25] introduced a monotonic improvement guarantee for MARL by

⁵We have thoroughly investigated the design of MARL training scheduler. Given that the data volume of task completion and cumulative rewards during early or even entire MARL training stages is typically insufficient to train converged and stable ML-based (including RL-based) Difficulty Schedulers. Furthermore, the mutual dynamic influence between MARL and the training scheduler’s learning process may lead to unstable training conditions. Compared to these ML-based strategies, statistical methods with parameters like window size and thresholds possess clear physical meanings and offer more interpretable logic than machine learning black-box models. More importantly, statistical-based training scheduler, such as linear regression, features low computational complexity, enabling stable and effective environment difficulty adjustment without additional training overhead.

enforcing trust region constraints over joint policies, providing a theoretical foundation for proximal ratio clipping.

To better address the *Credit Assignment Problems* in the *Centralized Training with Decentralized Execution (CTDE)* paradigm, QMIX [19] optimizes VDN [26] by introducing a hypernetwork that combines local value functions with global state information, replacing the simple summation of local Q-values in VDN. Foerster et al. [18] proposed COMA, which employs a centralized critic with a counterfactual advantage function that marginalizes each agent's action while holding others fixed, enabling precise credit assignment in MARL.

For the efficient experience utilization, Yang et al. [39] proposed MARR, a novel framework that enhances experience utilization efficiency in multiagent systems through a reset strategy and data augmentation technique, significantly improving sample efficiency in MARL. Moreover, several studies enhanced learning experience efficiency in RL by incorporating episodic memory buffers [9], [10]. These approaches record optimal state-action Q-values from historical trajectories, serving as retrievable memory signals to enhance learning from experience buffers.

In addition to the aforementioned approaches, recent works tackled existing problems such as partial observability and complex environmental dynamics through enhanced inter-agent communication mechanisms [29]–[31] that facilitate efficient information sharing, as well as developing innovative agent grouping strategies [28], [40] employing hierarchical architectures to effectively integrate localized observations.

In cooperative-adversarial MARL scenarios, the most widely adopted benchmark environments include SMAC [14], [15], google research football (GRF) [41], Pommerman [42], and Neural MMO [43]. Empirical reviews indicated that approximately 72% of recent MARL publications utilize SMAC, while nearly 58% employ GRF as primary testbeds [44]–[46]. Among them, SMAC [14], based on *StarCraft-II*, *Blizzard Entertainment*TM, has become the benchmark environment for evaluating multi-agent cooperative-adversarial algorithms due to its 40+ heterogeneous battle scenarios and flexible adversarial mechanics. By featuring diverse unit compositions and adjustable AI difficulty levels, it provides an ideal testing platform for MARL policy development.

Despite significant progress in MARL achieved through both algorithmic innovations and benchmark development [1], [7], [46], a fundamental limitation still persists: “*environmental meta-stationarity*”. In other words, most current MARL methods are evaluated under fixed-difficulty adversarial conditions (e.g., SMAC’s default Level-7 AI opponents). Such meta-stationary conditions often leads agents to develop environment-specific strategies that may result in environment-overfitted policies with poor generalization and convergence to locally optimal but globally suboptimal strategies. Hence, the primary objective of this work is to systematically integrate CL [16] into cooperative-adversarial MARL, addressing the long-overlooked bottleneck of “*environmental meta-stationarity*”.

III. PRELIMINARIES

A. Multi-Agent Reinforcement Learning

MARL is essentially a sequential decision-making task under multi-agent cooperative context, which can be formally modeled as a *decentralized partially observable markov decision process (Dec-POMDP)* [47]. Formally, a Dec-POMDP is defined by the tuple $G = \langle \mathcal{S}, \mathcal{U}, P, r, \Omega, O, n, \gamma \rangle$, where $\mathcal{S}$ is the state space, $\mathcal{U}$ is the action space for each of the n agents, $P(s'|s, \mathbf{u}) : \mathcal{S} \times \mathcal{U}^n \times \mathcal{S} \rightarrow [0, 1]$ is the state transition function, $r(s, \mathbf{u}) : \mathcal{S} \times \mathcal{U}^n \rightarrow \mathbb{R}$ is the shared reward function, Ω is the observation space with emission probabilities $O(s, a) : \mathcal{S} \times \mathcal{A} \rightarrow \Omega$, and $\gamma \in [0, 1]$ is the discount factor. At each timestep t , each agent $a \in \{1, \dots, n\}$ selects an action $u^a \in \mathcal{U}$, forming a joint action $\mathbf{u} \in \mathcal{U}^n$. The environment transitions according to $P(s_{t+1}|s_t, \mathbf{u}_t)$, generating a scalar reward $r_t = r(s_t, \mathbf{u}_t)$. Agents receive individual observations $z_t^a \sim O(s_t, a)$ that may represent partial state information.

Each agent maintains an action-observation history $\tau_t^a \in \mathcal{T} = (\Omega \times \mathcal{U})^*$, which maps to actions through a stochastic policy $\pi^a(u^a|\tau^a) : \mathcal{T} \times \mathcal{U} \rightarrow [0, 1]$. The joint policy π induces a centralized action-value function:

$$Q^\pi(s_t, \mathbf{u}_t) = \mathbb{E}_{s_{t+1:\infty}, \mathbf{u}_{t+1:\infty}} \left[\sum_{i=0}^{\infty} \gamma^i r_{t+i} \mid s_t, \mathbf{u}_t \right]. \quad (1)$$

While training can leverage centralized information (full state s and all agents' histories τ), execution must be decentralized - each agent's policy π^a depends only on its local history τ^a . This framework subsumes both the fully observable MMDP case (when $O(s, a) = s$) and standard POMDPs (when $n = 1$). The key challenge emerges from the exponential growth of joint action space $\mathcal{U}^n$ and the partial observability constraints during execution.

MARL algorithms are typically categorized into three training paradigms: *centralized training and execution (CTCE)* [48]–[50], *decentralized training and execution (DTDE)* [51], and *centralized training with decentralized execution (CTDE)* [52], [53]. Among these, CTDE has become predominant due to its superior learning efficiency while maintaining the flexibility of decentralized execution. The algorithms under CTDE paradigm combine decentralized execution where each agent a acts via its own policy $\pi^a(u^a|\tau^a)$ conditioned on local action-observation histories $\tau^a = (z_0^a, u_0^a, \dots, z_t^a)$, with centralized training that optimizes a joint action-value function $Q_{tot}(\tau, \mathbf{u}, s) = g(\{Q^a(\tau^a, u^a)\}_{a=1}^n, h(s))$ where: (1) individual utilities Q^a depend only on local τ^a , (2) the state encoder $h(s)$ integrates global information, and (3) the mixing network $g(\cdot)$ enforces consistency between decentralized policies and centralized value estimation through the monotonicity constraint $\partial Q_{tot} / \partial Q^a \geq 0$. Notable algorithms including VDN [26], QMIX [19], MAPPO [54], and MADDPG [55] all adhere to the CTDE paradigm. Similarly, our proposed method is also based on the baseline algorithms of CTDE.

B. Curriculum Learning (CL)

CL [16] is a novel continuous learning paradigm originally proposed by Bengio et al. [56] for supervised learning tasks in

CV and NLP. This paradigm progressively exposes a machine learning model to data samples of increasing complexity, mimicking the structured learning process in human education. Formally, in supervised training, given a training dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{|\mathcal{D}|}$ where $x_i \in \mathcal{X}$ is an input (e.g., an image or text sequence), $y_i \in \mathcal{Y}$ is its corresponding label, and $|\mathcal{D}|$ denotes the sample size, CL introduces a *difficulty measure* $d : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}^+$ that quantifies the complexity of each training example.

The learning process is governed by a *curriculum scheduler*, or *training scheduler*, $\mathcal{C}(t)$, which defines a sequence of training distributions $\{Q_t\}_{t=1}^T$ over $\mathcal{D}$, let T denote the number of curriculum stages. At each training stage t , the model optimizes a reweighted objective: $\min_{\theta} \mathbb{E}_{(x,y) \sim Q_t} [\mathcal{L}(f_{\theta}(x), y)]$, where f_{θ} is the model with parameters θ , $\mathcal{L}$ is the loss function, and Q_t is a distribution that prioritizes easier examples early in training before gradually incorporating harder ones. The scheduler ensures that the effective training data transitions smoothly from a simpler subset $\mathcal{D}_1 \subset \mathcal{D}$ to the full dataset $\mathcal{D}_T = \mathcal{D}$, satisfying:

$$\mathbb{E}_{Q_t}[d(x, y)] \leq \mathbb{E}_{Q_{t+1}}[d(x, y)], \quad \forall t \in \{1, \dots, T-1\}. \quad (2)$$

Formally, the implementation of *curriculum scheduler* can be defined Q_t as a reweighting of the original data distribution $P(x, y)$: $Q_t(x, y) = w_t(x, y) \cdot P(x, y)$, where $w_t(x, y)$ is a time-dependent weight function that increases for harder examples as t progresses. For instance, in *self-paced learning* [57], a CL variant, its weights are binary ($w_t \in \{0, 1\}$) and determined by a threshold τ_t on difficulty: $w_t(x, y) = \mathbb{I}[d(x, y) \leq \tau_t]$, with $\tau_t \leq \tau_{t+1}$. Noteworthy, the final effectiveness of curriculum is determined by the alignment between $d(x, y)$ and the model's current capability—examples deemed ‘easy’ should be those the model can learn reliably at stage t , while ‘hard’ examples require more mature representations. In NLP, $d(x, y)$ might measure sentence length or syntactic complexity. In CV, it could reflect image clutter, label noise, or distributional mismatch. The scheduler $\mathcal{C}(t)$ may be predefined (e.g., linear annealing of τ_t) or adaptively tuned based on model performance.

IV. PROGRESSIVE CL FOR MARL

In this section, we propose an innovative algorithmic innovation designed to enhance MARL baseline methods through CL. We refer to this approach as the *CL-enhanced MARL framework* (CL-MARL). As illustrated in Fig. 2, our approach integrates two novel components, i.e., **FlexDiff** and **CGRPA**. The framework addresses two critical challenges, i.e., the inherent limitations of “*environmental meta-stationarity*” in conventional MARL paradigms, and the *training instability* caused by abrupt environmental difficulty transitions in CL.

A. Flexible Difficulty Scheduler

The fundamental challenges in CL lie in the implementation of the *difficulty measurer* and the determinization of the *training scheduler*.

Algorithm 1 The Difficulty Scheduling Process of **FlexDiff**

Require: Observation window $\mathcal{W}_t$, current difficulty d_t , momentum m_{t-1} ;

- 1: Update running statistics μ_w, σ_w^2 (Eq. 4)
- 2: Compute trend slope β_w via normal equations (Eq. 7)
- 3: Derive reward convexity $\nabla^2 r_t$ from buffer (Eq. 8)
- 4: Update momentum m_t (Eq. 9)
- 5: Calculate adaptive thresholds $\tau_h(d_t), \tau_l(d_t)$
- 6: Difficulty Change with Eq. 12;
- 7: **if** $\mu_w > \tau_h$ **and** stability $\mathcal{S} = 1$ **and** momentum $m_t > 0.2$ **then**
- 8: Increase difficulty: $d_{t+1} \leftarrow \min(d_t + \delta_+, D_{max})$
- 9: **else if** Performance collapse detected ($\mu_w < \tau_l$ **or** $\nabla^2 r_t < -0.05$) **and** $m_t < -0.2$ **then**
- 10: Decrease difficulty: $d_{t+1} \leftarrow \max(d_t - \delta_-, 1)$
- 11: **end if**
- 12: **return** Adjusted difficulty d_{t+1} ;

1) The Implementation of Difficulty Measurer for SMAC:

In CL for supervised tasks such as CV and NLP, the difficulty measurer partitions data into levels based on sample complexity [16]. For CL-based MARL, difficulty is defined through task-level variations, commonly by adjusting environment parameters. For SMAC, this involves modifying the opponent AI difficulty via the StarCraft II scripting interface.

Rather than requiring pre-configuration within the StarCraft II engine, the *difficulty measurer* in SMAC leverages a built-in API to directly adjust opponent AI levels during training. This enables flexible and automated difficulty modulation without manual scripting. As illustrated in Part 2 of Fig. 2, the API provides 10 discrete difficulty levels. The built-in AI opponents difficulty levels are: 1 (*Very Easy*), 2 (*Easy*), 3 (*Medium*), 4 (*Hard*), 5 (*Harder*), 6 (*Very Hard*), and 7 (*Elite*). Beyond these, levels 8–10 are “cheat-level” AIs that gain unfair advantages: 8 (*Map hack*), 9 (*Resource Cheating*), and 10 (*Map hack + Resource Cheating*), where level 10 approximates the skill of top 30–50% human players [58]. This API-based design makes the difficulty measurer both efficient and adaptable for curriculum-based MARL training. For details, refer to Blizzard’s official API documentation or in-game AI settings⁶.

2) *The Implementation of FlexDiff*: Typically, the training scheduler under supervised learning setting determines when and how to introduce harder samples for training [16]. Correspondingly, the training scheduler in MARL context depends when to increase or decrease overall environmental difficulty, i.e., opponents’ strength in multi-agent adversarial scenarios, where evident signals during agents training serve as criteria for task difficulty adjustment.

Hence, our proposed adaptive training difficulty scheduler, i.e., **FlexDiff**, dynamically modulates environmental difficulty based on real-time agent performance, primarily relying on the real-time obtained rewards and task completion rates (e.g., SMAC average rewards and win rates).

⁶Blizzard’s official API: <https://develop.battle.net/documentation>.

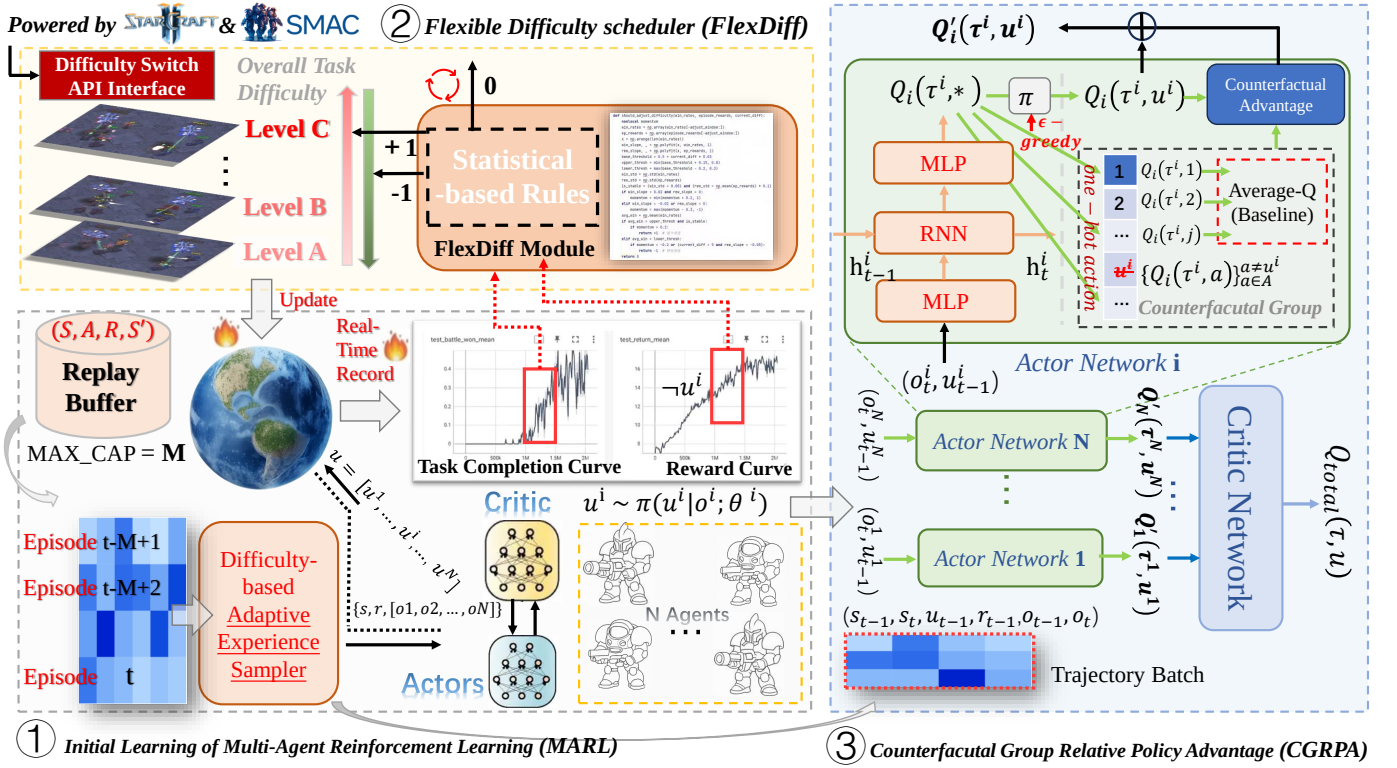


Fig. 2: The architecture of our proposed CL-based MARL framework for cooperative-adversarial scenarios in SMAC. The framework comprises: (1) the original MARL training with static environment settings, (2) our proposed **FlexDiff** scheduler for dynamic difficulty adjustment, and (3) our introduced CGRPA algorithm enhancing *Actor-Critic* with counterfactual interaction advantage estimation.

As illustrated in Part 2 of Fig. 2, **FlexDiff** achieves self-adaptive CL through analyzing multi-scale statistical feature, with three key innovations: *synergistic dual-metric evaluation*, *momentum-driven dynamic adjustment buffer*, and *dynamic boundary constraints* for training stability.

As shown in **Algorithm 1**, **FlexDiff** establishes a dynamic mapping relationship between agent performance and environmental difficulty by monitoring dual-channel metrics (i.e., *Win Rate* and *Episode Reward*) in real-time during MARL training. This ensures the multi-dimensional model learning feedback during MARL policy optimization. We formalize its core mechanisms as follows:

Synergistic dual-metric performance evaluation: During the training phase, we conduct periodic evaluations at fixed intervals to assess the current model's decision-making capabilities. For each evaluation, we run multiple, e.g., 32 test episodes at the default difficulty level and calculate two key metrics, i.e., the average win rate and the mean cumulative reward. All historical tested performances are maintained in an experience buffer-like history list.

To monitor the agents' recent training progress, we employ two fixed-length sliding windows to track the performance metrics of win rates and averaged rewards. Formally, we define the performance evaluation window $\mathcal{W}_t$ at training step t as:

$$\mathcal{W}_t = \{(w_i, r_i)\}_{i=t-N+1}^t, \quad (3)$$

where $w_i \in [0, 1]$ represents the normalized win rate (victories

divided by total testing episodes) at the i_{th} evaluation cycle, and $r_i \in \mathbb{R}^+$ denotes the correspondingly episode reward.

The overall stability of agents' performance is formalized through the following constraints:

$$\sigma_w^2 = \frac{1}{N} \sum_{i=1}^N (w_i - \mu_w)^2, \quad \mu_w = \frac{1}{N} \sum_{i=1}^N w_i, \quad (4)$$

$$\sigma_r^2 = \frac{1}{N} \sum_{i=1}^N (r_i - \mu_r)^2, \quad \mu_r = \frac{1}{N} \sum_{i=1}^N r_i, \quad (5)$$

$$\mathcal{S} = \mathbb{I}(\sigma_w < 0.08) \otimes \mathbb{I}(\sigma_r < 0.1\mu_r), \quad (6)$$

where $\mathbb{I}$ acts as a binary classifier for stability conditions, $A \otimes B$ denotes the simultaneous satisfaction of both constraints A and B . The stability indicator $\mathcal{S} \in \{0, 1\}$ combines low value in both success rates and rewards, ensuring reliable policy evaluation before difficulty adjustments.

For the windowed win rates w_i , $i = 1, \dots, N$, the temporal trend β_w is quantified through linear regression:

$$\beta_w = \frac{N \sum x_i w_i - \sum x_i \sum w_i}{N \sum x_i^2 - (\sum x_i)^2}, \quad x_i = i - t + N - 1, \quad (7)$$

In Eq. (7), the slope parameter β_w reveals three distinct operational regimes: 1). significant improvement ($\beta_w > \zeta$), 2). stabilized performance ($|\beta_w| \leq \zeta$), and 3). performance degradation ($\beta_w < -\zeta$), where we define a threshold hyperparameter ζ to separate the upper and lower bounds of perfor-

mance fluctuations. The term $\sum$ denotes the summation across data points in the sliding window $1, \dots, N$. This enables early detection of learning convergences and breakthrough moments. Then, we compute second-order differences for reward to detect phase transitions:

$$\nabla^2 r_t = r_t - 2r_{t-1} + r_{t-2}. \quad (8)$$

In Eq. (8), positive values ($\nabla^2 r_t > 0$) indicate convex acceleration where marginal improvements grow exponentially, characteristic of effective policy refinement. Near-zero values ($\nabla^2 r_t \approx 0$) indicate linear progress with a constant learning rate, while negative values ($\nabla^2 r_t < 0$) signal a slowdown in convergence, where returns decrease, often preceding a performance collapse.

Momentum-driven dynamic adjustment buffer: To avoid frequent adjustments, we need to consider adding a cooling mechanism. Specifically, we introduce a self-adaptive momentum term $m_t \in [-1, 1]$ that integrates historical gradient information, which can be formalized as follows:

$$m_t = \gamma m_{t-1} + (1 - \gamma) \tanh(\beta_w + 0.5 \nabla^2 r_t), \quad (9)$$

with $\gamma = 0.8$ controlling memory decay. The hyperbolic tangent non-linearity bounds momentum growth while preserving directional sensitivity to performance trends.

Dynamic boundary constraints: To prevent the difficulty from being switched due to unreasonable win rate performance, which could lead to training instability, we employ a flexible threshold mechanism that adapts to the current difficulty level d_t , further constraining the difference scheduler:

$$\tau_h(d_t) = \min(\gamma, 0.5 + 0.03d_t), \quad (10)$$

$$\tau_l(d_t) = \max(\eta, 0.4 - 0.02d_t). \quad (11)$$

This implements adaptive boundary self-regulation via the reason of higher difficulty levels require stronger evidence for progression (τ_h increases slower than τ_l decreases), enforcing cautious performance increasing and quick intervention to prevent performance decreasing.

Finally, as depicted in Eq. (12), the difficulty adjustment policy integrates multiple performance indicators through a three-branch decision structure:

$$d_{t+1} = \begin{cases} d_t + 1 & \text{if } \mu_w > \tau_h \wedge \mathcal{S} = 1 \wedge m_t > 0.2, \\ d_t - 1 & \text{if } (\mu_w < \tau_l \vee \nabla^2 r_t < -0.05) \wedge m_t < -0.2, \\ d_t & \text{otherwise.} \end{cases} \quad (12)$$

First, the step-by-step difficulty changes (± 1) ensure gradual progression matching standard difficulty levels. Second, the strict condition of task difficulty upgrade ($\mu_w \wedge \mathcal{S} \wedge m_t$) requires concurrent high performance, stability, and positive momentum. Third, the degradation trigger ($\mu_w \vee \nabla^2 r_t$) responds quickly to either direct failure or accelerating decline. On the above basis, this flexible mechanism prevents unstable training while maintaining learning progress.

B. Counterfactual Group Relative Policy Advantage (CGRPA)

1) Problem and Motivation: Consider a cooperative multi-agent system with N agents modeled as a Dec-POMDP. Let $Q_{\text{tot}}(\tau, \mathbf{u}|s)$ denote the joint action-value function conditioned

on the state s , where $\tau = (\tau_1, \dots, \tau_N)$ are agents' action-observation histories and $\mathbf{u} = (u_1, \dots, u_N)$ the joint action. The key challenge lies in *Asymmetric Credit Assignment*, $\partial Q_{\text{tot}} / \partial Q_i \geq 0$, in *actor-critic* (AC)-based MARL methods (such as QMIX constraint), which could lead to sub-optimal dilemma. Traditional value-based MARL algorithms [2], [46] often struggle with policy convergence and recovery after environmental difficulty changes, especially when CL technique is applied to dynamically adjust the complexity of the task.

In light of these challenges, we introduce a novel algorithm, **CGRPA**, to enhance the agents' robustness during curriculum difficulty switching. **CGRPA** rethinks how credit is assigned to each agent's actions by comparing their actual actions with potential counterfactual actions. This allows agents to rapidly adapt and adjust their strategies after changes in task difficulty introduced by CL.

2) Action-Group Counterfactual Advantage: As shown in Fig. 3, **CGRPA** reconstructs the advantage function and introduces a innovative action-group relative advantage estimation, through dual mechanisms: *counterfactual reasoning* [18] and *group relative policy optimization* [17] to achieve precise per-agent credit assignment.

During Q-value estimation in MARL training, the counterfactual advantage can be formalized as Eq. (13):

$$A_i^{\text{CF}}(s, \mathbf{u}) = \underbrace{Q_{\text{tot}}(s, \mathbf{u}) - \mathbb{E}_{\bar{\mathbf{u}}_i \sim \pi_i}[Q_{\text{tot}}(s, (\mathbf{u}^{-i}, \bar{\mathbf{u}}_i))]}_{\text{Counterfactual Baseline}} - \alpha D_{\text{KL}}(\pi_i \| \bar{\pi}_g), \quad (13)$$

where $\mathbf{u}^{-i}$ denotes the joint actions excluding i -th agent's action, and α is temperature parameter controlling coordination strength⁷. The first term quantifies individual contribution by comparing the actual action u_i with its policy's averaged expectation value, while the *KL divergence* term, D_{KL} , constrains policy deviation from the group average. The temperature parameter α balances these components. This design breaks traditional valuable-based MARL methods' (such as VDN, QMIX) monotonicity constraint, enabling asymmetric and effective credit assignment.

3) Group-Relative Policy Optimization: The policy update rule combines counterfactual credit assignment with group relative policy:

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\mathcal{D}} \left[\sum_{i=1}^N A_i^{\text{CF}}(s, \mathbf{u}) \nabla_{\theta} \log \pi_i(u_i | \tau_i) \right] \quad (14)$$

$$\text{s.t. } Q_{\text{tot}} = \text{Critic}(Q_1, \dots, Q_N; \psi), \quad \psi \in \Psi_{\text{mono}}. \quad (15)$$

where $\nabla_{\theta} J(\theta)$ denotes the optimization target of counterfactual relative advantage, $\bar{\pi}_g(u|s) = \frac{1}{N} \sum_{j=1}^N \pi_j(u | \tau_j)$ represents the group relative policy.

Typically, there are two primary distinctions from conventional AC-based MARL frameworks. First, during execution, each π_i relies solely on local history τ_i , thereby preserving decentralization. Second, the training process utilizes a critic network (e.g., the mixing network in QMIX [19]), denoted

⁷The KL divergence D_{KL} in Eq. (13) creates a potential game where agents are rewarded for aligning with $\bar{\pi}_g$.

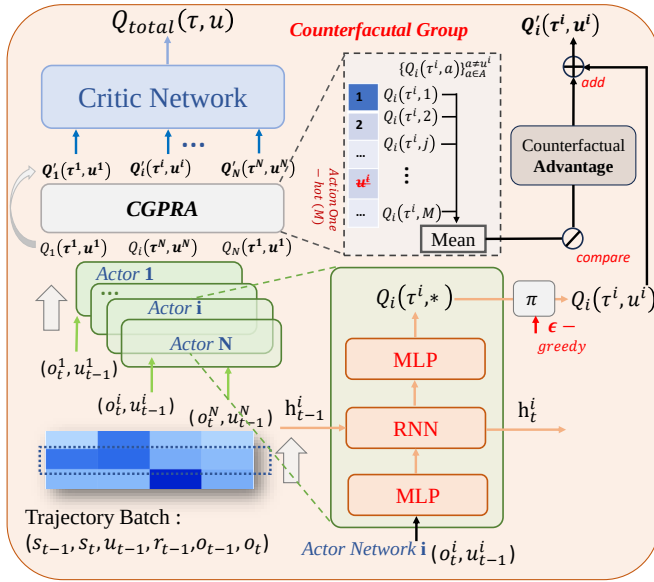


Fig. 3: The detailed visualization of our proposed CGRPA algorithm, corresponding to the Part 3 of Fig. 2.

as Critic($\cdot; \psi$), which theoretically preserves monotonicity while actively reducing projection bias through counterfactual advantage-weighted learning.

Specifically, the A^{CF} term in Eq. (13) guides gradient updates through two mechanisms. The counterfactual component enables more effective individual contribution assessment, while the KL divergence maintaining policy consistency.

4) *Q-Function Update with Counterfactual Infusion:* We incorporate the counterfactual advantages $A_i^{\text{CF}}(s, \mathbf{u})$ into individual Q-value estimation:

$$\tilde{Q}_i(\tau_i, u_i) = Q_i(\tau_i, u_i) + \lambda A_i^{\text{CF}}(s, \mathbf{u}), \quad (16)$$

$$Q_{\text{tot}} = f_{\psi} \left(\{\tilde{Q}_i\}_{i=1}^N, s \right), \quad (17)$$

where $\lambda \in (0, 1]$ controls counterfactual integration strength, A_i^{CF} is the counterfactual advantage from Eq. (13), and f_ψ maintains monotonicity such that $\partial f_\psi / \partial \tilde{Q}_i \geq 0$.

The modified TD-target computation becomes:

$$y = r + \gamma f_{\psi^-}(\{Q_i^-(\tau'_i, u'^*_i) + \lambda A_i^{\text{CF}}(s', \mathbf{u}'^*)\}, s'), \quad (18)$$

with target network parameters ψ^- and Q_i^- . The complete loss function is:

$$\mathcal{L}(\theta) = \mathbb{E}_{\mathcal{D}} \left[\left(f_{\psi}(\{\tilde{Q}_i\}, s) - y \right)^2 \right] + \beta \sum_{i=1}^N D_{\text{KL}}(\pi_i \| \bar{\pi}_g). \quad (19)$$

For simplicity, Algorithm 2 outlines the key steps of our **CGRPA** method. Overall, **CGRPA** improves credit assignment by comparing each agent’s actual actions with potential counterfactual actions, allowing agents to quickly adapt their strategies to the task complexity variations introduced by CL.

V. EXPERIMENTS

This section validates CL-MARL through systematic experiments. Section V-A details the setup, including the SMAC

Algorithm 2 Implementation Details of CGRPA

Require: Learning rate η , discount factor γ , critic network

$$\begin{aligned}
& f_{\psi} \\
1: & \text{Initialize: Agent networks } \{Q_i, \pi_i\}_{i=1}^N \text{ with parameters } \theta, \text{ Target networks } \{Q_i^-, \pi_i^-\}_{i=1}^N \leftarrow \{Q_i, \pi_i\}, \text{ Critic} \\
& \text{networks } f_{\psi} \text{ and target } f_{\psi^-} \leftarrow f_{\psi}, \text{ Replay buffer } \mathcal{D} \leftarrow \emptyset. \\
2: & \textbf{for} \text{ episode} = 1 \textbf{ to } M \textbf{ do} \\
3: & \quad \text{Collect trajectory } \{(s_t, \tau_t, \mathbf{u}_t, r_t)\} \text{ using } \epsilon\text{-greedy}; \\
4: & \quad \textbf{for} \text{ gradient step} = 1 \textbf{ to } K \textbf{ do} \\
5: & \quad \quad \text{Sample mini-batch } B \sim \mathcal{D}; \\
6: & \quad \quad \text{Compute counterfactual advantage } A_i^{\text{CF}}; \text{ (Eq. 13)} \\
7: & \quad \quad \text{Compute new Q-values for each } i\text{-th agent, where} \\
& \quad \quad i \in \{1, \dots, N\}; \text{ (Eq. 16):}
\end{aligned}$$

$$\tilde{Q}_i = Q_i(\tau_i, u_i) + \lambda A_i^{\text{CF}}$$

8: Calculate current Q_{tot} : $Q_{tot} = f_{\psi} \left(\{\tilde{Q}_i\}, s \right)$;

9: Compute target actions u'^* :

$$\mathbf{u}'^* = \operatorname{argmax}_{\mathbf{u}'} f_{\psi-}(\{Q_i^-(\tau'_i, u'_i)\}, s');$$

10: Calculate target value:

$$y = r + \gamma \cdot f_{\psi-}(\{Q_i^-(\tau'_i, u_i'^*) + \lambda A_i^{\text{CF}}\}, s');$$

11: Update mixing network via:

$$\psi \leftarrow \psi - \eta \nabla_{\psi} \frac{1}{|B|} \sum_{k=1}^{|B|} (Q_{\text{tot}}^k - y^k)^2;$$

12: Update agent policies with coordination regularization: $\theta_i \leftarrow \theta_i - \eta \nabla_{\theta_i} \left[A_i^{\text{CF}} \log \pi_i(u_i | \tau_i) + \alpha D_{\text{KL}}(\pi_i \| \bar{\pi}_g) \right] \quad \forall \text{ agent } i.$

13: **end for**14: **end for**

benchmark, hyperparameters, and hardware. Section V-B compares CL-MARL with baselines across SMAC maps of varying difficulty. Section V-C presents ablation studies on component importance, performance gains, and theoretical support.

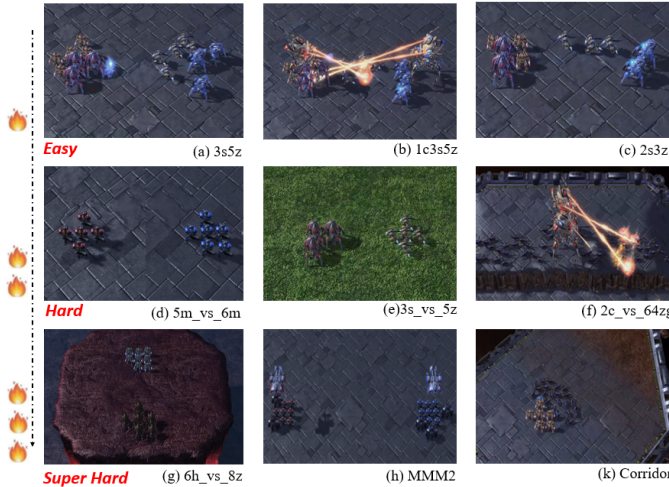
A. Experimental Setup

This section details the experimental setup, including: the adopted SMAC environment, model hyperparameter configurations, experimental procedure designing, and hardware implementation specifications.

1) *Decentralised StarCraft II Micromangement*: The SMAC environment is a decentralized MARL benchmark based on *StarCraft II* micromangement [14]. In SMAC, each RL agent controls an individual unit (Marines, Stalkers, etc.) with partial observability [local unit positions, health] and discrete actions [move, attack, stop]. On the allied side, each unit is independently controlled by decentralized learning agents, contrasting with enemy units that are managed by *StarCraft II*'s native AI utilizing predefined heuristic rules. The action space of SMAC agents is discrete, comprising *movement* [4 directions], *attack* [if target is in range], *stop*, and *no-op* commands. Each agent receives partial

TABLE I: Details of the adopted SMAC experimental scenarios and maps, covering Easy, Hard, and Super-Hard levels.

Map Name	Difficulty	State Dim	Action Dim	Ep. Len
3m	Easy	90	13	60
5m	Easy	135	15	120
8m	Easy	192	18	120
2s_3z	Easy	130	12	120
1c_3s_5z	Easy	270	15	180
3s_5z	Easy	216	14	150
5s10z	Hard	380	25	200
7s7z	Hard	350	24	200
5m_vs_6m	Hard	98	12	70
8m_vs_9m	Hard	154	14	120
3s_vs_5z	Hard	68	11	250
2c_vs_64zg	Hard	396	65	600
bane_vs_bane	Hard	288	25	200
1c3s8z_vs_1c3s9z	Hard	425	33	180
MMM2	Super Hard	322	18	180
3s5z_vs_3s6z	Super Hard	230	15	170
27m_vs_30m	Super Hard	342	31	180

Fig. 4: The typical SMAC benchmark scenarios and maps with progressive difficulty levels, ranging from *Easy*, *Hard* to *Super-Hard*, for MARL algorithm testing.

observations including relative positions, distances, types, and status (health, shields) of units within its sight range. During training, a global state (hidden during testing) provides full information about all units' coordinates, health, cooldowns, and allies' previous actions. The detailed configurations of typical SMAC cooperative scenarios across different difficulty levels are provided in Table I. In this table, *State Dim* representing the state space dimension, *Action Dim* indicating the action space dimension, and *Ep. Len* denotes the episode length measured in time steps. Notably, Super Hard scenarios like *3s5z_vs_3s6z* and *corridor* remain unsolved by SOTA methods, such as QMIX [19], MARR [11], DER [20], EMC [9] and EMU [10], establishing SMAC as a key benchmark for cooperative MARL research.

SMACv2⁸ [15] enhances the original SMAC benchmark

⁸SMACv2 Code: <https://github.com/oxwhirl/smacv2>.

TABLE II: Experimental Hyperparameter Settings for our CL-MARL Framework within QMIXBackbone

Parameters. ori-QMIX	Value	Parameters. CL-MARL	Value
Learning rate (α)	5e-4	Maximum Difficulty	10
Gradient norm clip	10	Minimum Difficulty	1
Discount factor (γ)	0.99	Default Difficulty	7
RNN hidden dim	64	Start Difficulty	5
QMIX hidden dim	32	sliding window length	20
Traning Batch size	32	win-rate increase limit	0.08
Replay buffer capacity	5,000	reward increase limit	0.1
Target update interval	200	momentum threshold ζ	0.1
initial ϵ	1.0	max win-rate γ (FlexDiff)	0.75
minimized ϵ	0.05	min win-rate η (FlexDiff)	0.25
ϵ -anneal steps	50,000	CGRPA Loss Weight λ	0.5

[14] through three initialization-phase modifications while preserving the fundamental map configurations, including: randomized unit starting positions (either *surrounded* or *reflected* configurations), randomized unit types per race (*Terran*, *Protoss*, and *Zerg*) with controlled spawn probabilities, and adjusted sight and attack ranges that align with official StarCraft II values. These modifications enhance scenario diversity and prevent algorithmic overfitting. Notably, the built-in opponent difficulty API in SMACv2 serves as an effective difficulty measurer in our CL-MARL, making SMACv2 particularly suitable for progressive training. Thus, the experiments in our work are entirely based on the enhanced *StarCraft-II unit micromanagement* benchmark, SMACv2.

To comprehensively validate the effectiveness of our proposed method, we conduct extensive experiments across most SMAC scenarios, spanning three difficulty levels: *Easy*, *Hard*, and *Super Hard*. Through these thorough evaluations, we demonstrate the universal superiority of our CL-MARL framework. Meanwhile, we list several representative scenarios from each difficulty level in Fig. 4.

2) *Architecture and Training Details*: Our proposed CL-MARL framework modifies the original QMIX [19] architecture while maintaining identical hyperparameter configurations to ensure the overall experimental rigor and comparability⁹. The hyperparameter configurations for the original QMIX backbone network and the newly introduced CL-MARL modules, are detailed in Table II. The full implementation details, including environment configuration, network architecture, and training procedures, can be found in our released project¹⁰.

The evaluation metric in SMAC is the *test-win-rate*, which is measured at different training stages¹¹. All experimental results, i.e., mean test-win-rates, reported in subsequent sections are computed over 5 independent runs for each method across selected maps, accompanied by 85% confidence intervals. The implementation of CL-MARL is based on PyTorch 1.12.1 and Python 3.8.5, running on Ubuntu 20.04.1. The hardware

⁹The source codes of QMIX and other MARL algorithms are available at <https://github.com/starry-sky6688/MARL-Algorithms>

¹⁰CL-MARL: <https://github.com/NICE-HKU/CL2MARL-SMAC>.

¹¹The win-rate is calculated by the fraction of test episodes in which the learning agents successfully destroy all enemy units within the time limit.

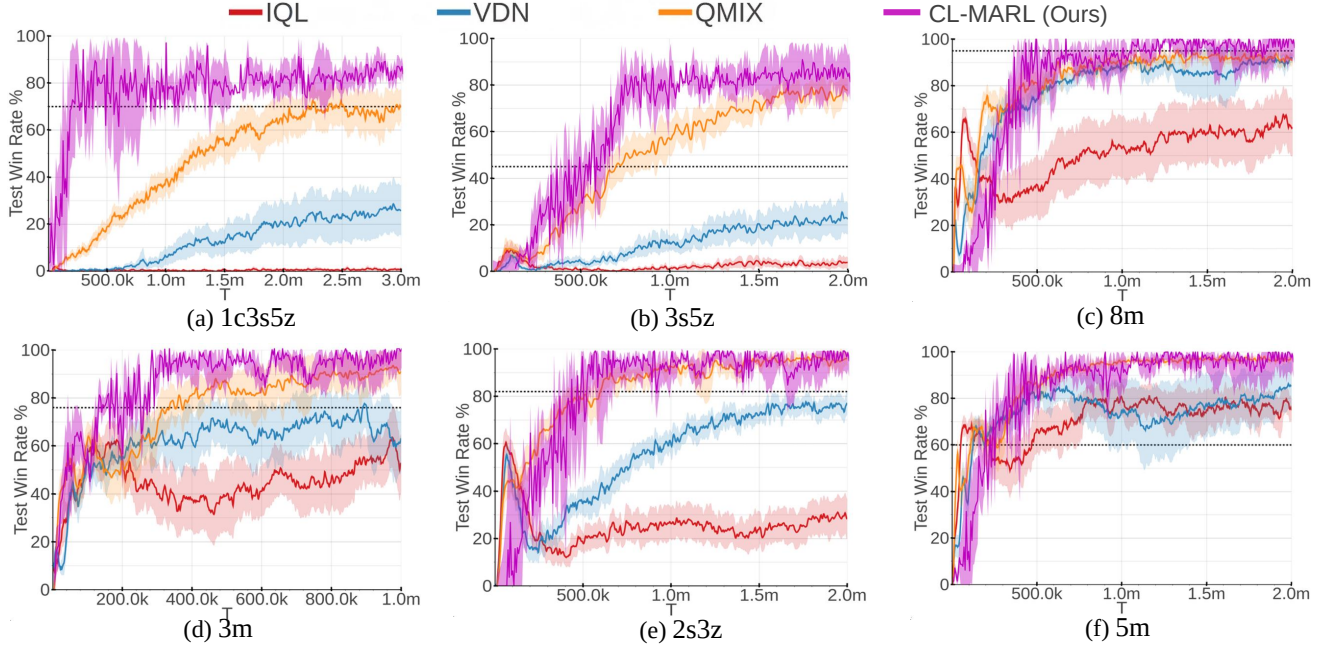


Fig. 5: Baseline Comparisons of the six easy maps in the SMAC benchmark. The x-axis represents the global training steps, while the y-axis records the average win rate progression of the RL algorithm.

platform consists of four Intel Xeon Platinum 8488C CPUs (48 cores each) and four NVIDIA GeForce RTX 4090 GPUs.

B. Main Comparison Results

In this section, we present extensive experimental results on the SMAC benchmark to validate the effectiveness and superiority of our proposed CL-MARL framework in the multi-agent cooperative-adversarial scenarios.

We systematically compare our method against SOTA MARL algorithms, including VDN [26], QMIX [19], OW-QMIX [8], QTRAN [27], QPLEX [59], MARR [11], DER [20], EMC [9], and EMU [10], across a diverse set of SMAC maps spanning three difficulty levels: *Easy*, *Hard*, and *Super Hard*. In this study, we evaluated our method on nearly 20 different SMAC benchmark maps, significantly more than in previous works, such as QMIX and OW-QMIX [8], [19] (6 maps each), DER [20] (4 maps), and MARR [11] (7 maps), EMU [10] (10 maps), EMC [9] (11 maps). To the best of our knowledge, our experiment represents one of the most comprehensive evaluations to date on SMAC benchmark, encompassing a wide variety of situations, from simple skirmishes to highly complex large-scale battles.

Based on the average task difficulty of maps tested, we organize our experimental comparisons into three parts: Sections V-B1, V-B2, and V-B3. Since our CL-MARL is built based on the original QMIX [19], the subsequent experimental analysis will focus more on comparisons between our model and QMIX as well as its related variants [8], [27], [59].

1) *Performance on Easy Maps*: First, we evaluate the learning capability of our proposed CL-MARL on six easy maps. As shown in Fig. 5, the test maps are: (a) 1c3s5z, (b) 3s5z, (c) 8m, (d) 3m, (e) 2s3z, and (f) 5m. The map names

indicate the number and types of units for both the learning agents and their opponents. For instance, 2s_vs_1sc denotes 2 *Stalkers* (controlled by RL agents) versus 1 *Spine Crawler* (controlled by the SC2 script), and 1c3s5z represents both adversarial sides having the unit composition, i.e., the agents of {1 *Colossus*, 3 *Stalkers*, and 5 *Zealots*}.

Across these maps, CL-MARL consistently demonstrates strong performance in learning optimal joint policies. Specifically, CL-MARL achieves a 100% win rate on four maps and demonstrates significantly faster policy exploration speed on the 3m, 3s5z, and 1c3s5z maps. Notably, in more complex maps like 1c3s5z and 3s5z, conventional algorithms such as IQL, VDN, and QMIX often stagnate in local optima, likely due to the static difficulty setting. In contrast, CL-MARL consistently avoids these suboptimal solutions by leveraging progressive difficulty adaptation. Furthermore, on maps like 2s3z and 8m, CL-MARL converges to the optimal policy notably earlier than other approaches. This improvement is attributed to the CL-based framework, which enables agents to acquire strategies in a structured, easy-to-hard progression and generalize more effectively across tasks.

While Fig. 5 highlights improvements in both win rate and convergence speed, we observe that CL-MARL occasionally exhibits less training stability than baseline algorithms (QMIX and VDN) during phases of rapid performance gain. This transient instability arises from the **FlexDiff** module’s dynamic adjustment of task difficulty, which may temporarily perturb performance consistency. However, this effect is limited and does not compromise final performance. More importantly, these observations validate that our proposed **CGRPA** module ensures robust policy adaptation during difficulty transitions, effectively mitigating instability and preserving long-term learning efficiency.

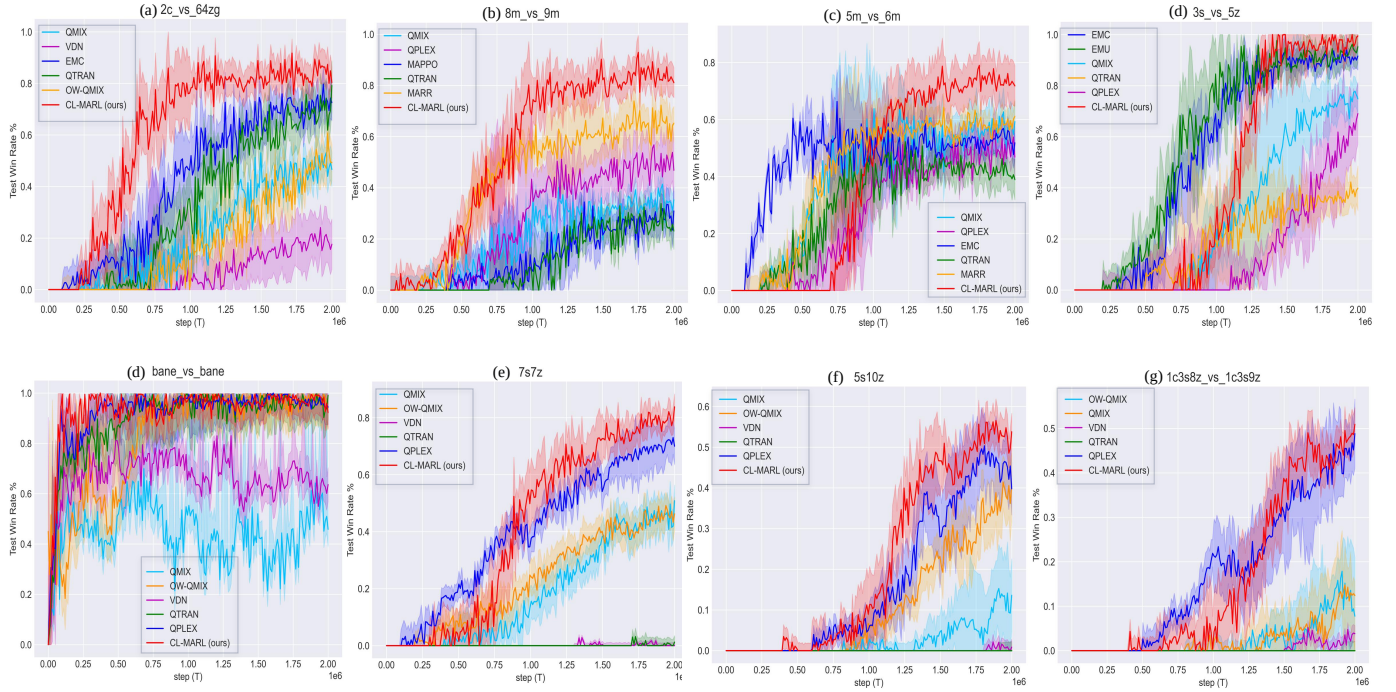


Fig. 6: Baseline Comparisons of the eight maps of *Hard Level* in the SMAC benchmark [x-axis: global training steps, y-axis: average win rate].

2) *Performance on Hard Maps:* Fig. 6 presents the experimental results of our CL-MARL on eight challenging maps. The results clearly demonstrate that our method achieves comparable performance with SOTA baseline algorithms in most scenarios, including *2c_vs_64zg*, *8m_vs_9m*, and *5m_vs_6m*. These findings further validate the effectiveness of our approach in high-difficulty environments. Notably, CL-MARL outperforms the current SOTA algorithm EMC (a curiosity-driven episodic memory variant of QPLEX) by over 8%-14% in win rate on the *2c_vs_64zg* map. Similarly, it surpasses another SOTA algorithm MARR by 10%-13% on both *8m_vs_9m* and *5m_vs_6m* maps.

As an enhanced variant of QMIX, CL-MARL significantly outperforms the original QMIX across all challenging maps while maintaining identical hyperparameters. For instance, on the *7s7z*, *5s10z*, and *1c3s8z_vs_1c3s9z* maps, the original QMIX achieves only approximately 45%, 10%, and 15% win rates after total 2 million training steps, respectively. In contrast, our CL-MARL demonstrates significant improvements, boosting the final win rates by approximately 20%, 40%, and 35% on these maps respectively. Moreover, CL-MARL maintains competitive performance compared to other SOTA algorithms including QTRAN, OW-QMIX, and QPLEX. These substantial gains underscore the effectiveness of our CL framework in enabling robust policy learning under difficult conditions, without requiring any parameter tuning.

Moreover, the results shown in Fig. 6 reveal that CL-MARL exhibits slower initial convergence speed compared to baselines on certain maps, including *7s7z*, *5m_vs_6m*, and *3s_vs_5z*. While baseline algorithms typically begin showing win rate improvements around 200,000 steps, CL-

MARL may require additional training steps over 500,000 steps during the early phase to adapt to varying environmental difficulties. This phenomenon primarily arises from the discrepancy between the initial training difficulty levels and the default testing difficulty settings. However, this temporary convergence delay does not compromise the final performance of our CL-MARL. The CL style in CL-MARL follows a progressive reinforcement process from simple to complex tasks. This design ensures robust learning in early stages, which ultimately contributes to superior capability in discovering optimal joint strategies. Overall, the initial learning stability trade-off is effectively compensated by the algorithm's enhanced final performance and better generalization ability.

3) *Performance on Super Hard Maps:* The experiments on three super hard SMAC maps (*27m_vs_30m*, *MMM2*, and *3s5z_vs_3s6z*) reveals distinct performance characteristics of CL-MARL under extreme challenges, as shown in Fig. 7.

In the *27m_vs_30m* map, CL-MARL achieves the highest win rate among all compared algorithms (including QMIX, QTRAN, and OW-QMIX). Notably, while QTRAN and OW-QMIX fail to achieve any meaningful win rate throughout training, our method maintains stable improvement and ultimately reaches a win rate of approximately 40%. This advantage can be attributed to the effective curriculum design of CL-MARL, which is particularly critical in large-scale combat settings involving 27 allied *Marines* against 30 enemy *Marines*. In contrast, the *MMM2* map presents a more complex challenge due to its heterogeneous unit composition. CL-MARL achieves comparable results to vanilla QMIX and OW-QMIX, but does not match the SOTA performance of QPLEX. This relative gap is attributed to the map's unique requirement

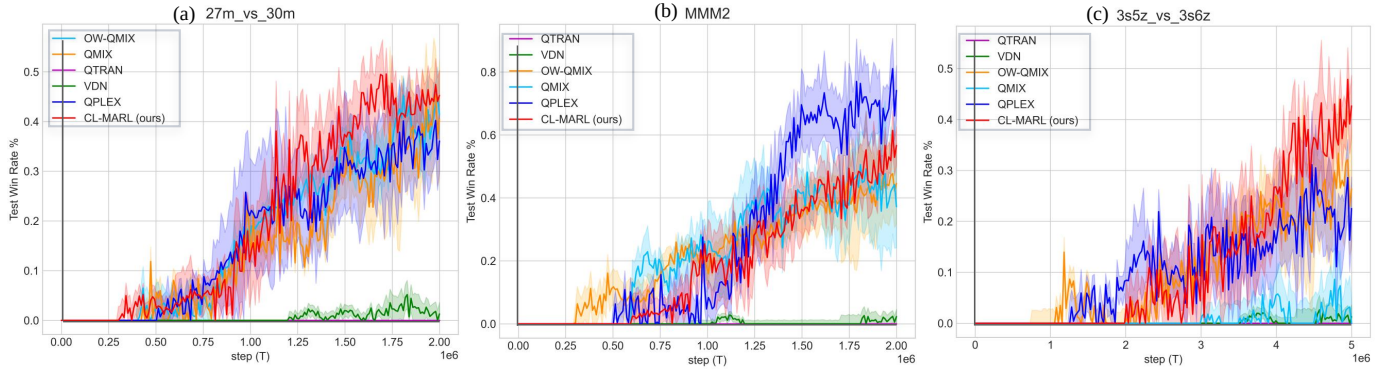


Fig. 7: Baseline Comparisons of the eight maps of *Super Hard Level* in the SMAC benchmark [*x-axis*: global training steps, *y-axis*: win rate].

for coordinated control of three distinct unit types with specialized roles. While our CL-based method successfully handles robust learning of general coordination, the current implementation may lack mechanisms for optimizing complex unit-type-specific micro-management. This limitation suggests an important direction for future improvement in handling heterogeneous unit compositions. In the *3s5z_vs_3s6z* scenario, CL-MARL demonstrates remarkable superiority over baseline methods. After 5 million training steps, our method achieves a 40% win rate, significantly surpassing over QMIX’s performance by approximate 30% and QPLEX’s results by 20%. This substantial improvement highlights our CL-MARL’s exceptional capability in handling asymmetric unit compositions. The progressive curriculum enables agents to first master fundamental *Stalker-Zealot* coordination before tackling the additional challenge of unit disadvantage (5 vs 6 *Zealots*).

Overall, these experimental findings collectively demonstrate that while CL-MARL does not consistently outperform all baselines across every super-hard environments, it provides substantial advantages in the most challenging scenarios where traditional methods often fail. The progressive learning approach embedded in our CL-based framework is especially effective in handling large unit counts, asymmetric matchups, and complex coordination requirements in multi-agent adversarial environments such as SMAC.

C. Ablated Studies

Here we provide detailed ablation studies to dissect the contributions of each component in CL-MARL.

1) *Mitigating FlexDiff’s Transition Instability with CGRPA*: To further investigate the underlying mechanisms of our CL-MARL, we conducted ablation experiments to validate the convergence risks induced by the **FlexDiff** module and the stabilization effects of **CGRPA**. Specifically, we removed the **CGRPA** module from the CL-MARL framework to observe its impact on policy convergence. As shown in Fig 8, the performance of CL-MARL (w/o CGRPA) was compared across three maps: *8m_vs_9m*, *bane_vs_bane*, and *2c_vs_64zg*.

On the *8m_vs_9m* map, the default CL-MARL exhibited continuous improvement, maintaining a high win rate throughout the training process. However, CL-MARL (w/o CGRPA)

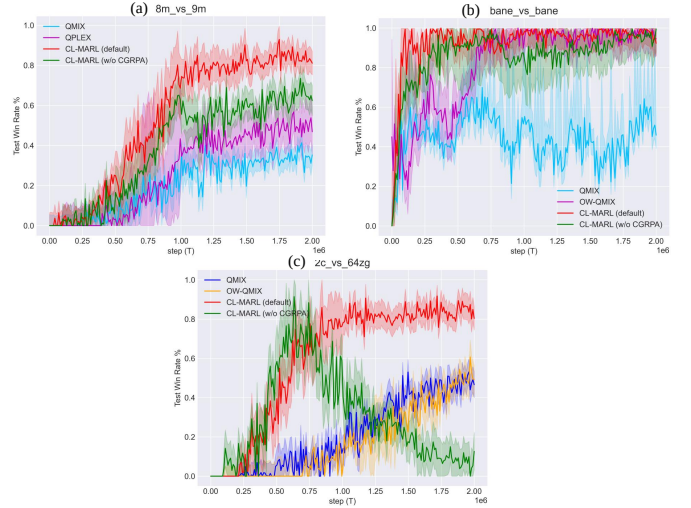


Fig. 8: Performance Comparison of Convergence Risks Induced by **FlexDiff** and Stabilization Effects of **CGRPA** on CL-MARL Learning.

shows a marked decline in win rate by approximately 10% at around 100k steps, with subsequent policy optimization slowing significantly. By the end of the training, the win rate stabilized at around 68%, far below the original CL-MARL’s 80% win rate. Similarly, on *bane_vs_bane*, when removing **CGRPA**, the model led to a significant drop in win rate at around 800k steps, with a slow recovery afterward. This demonstrates how the absence of **CGRPA** hinders the agents’ ability to recover from performance drops during training, thereby affecting long-term stability and convergence. In the *2c_vs_64zg* map, CL-MARL (w/o CGRPA) initially exhibited a similar win rate improvement trajectory as the original CL-MARL, with both showing rapid early improvement. However, around 600k steps, CL-MARL (w/o CGRPA) began to deteriorate as it failed to adapt to the difficulty adjustments introduced by the **FlexDiff** module. The win rate began to gradually decline, and the algorithm entered a suboptimal policy trap, with continued degradation triggered by dynamic task transitions. This clear divergence from the original CL-MARL highlights the crucial role of **CGRPA** in stabilizing

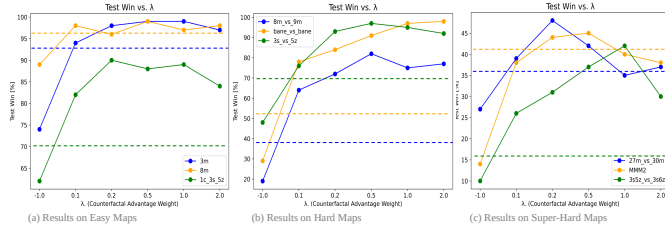


Fig. 9: Comparative Analysis on Normalized Win Rates of our CL-MARL Under Different Hyperparameter λ Values on Selected SMAC Maps

training stability under evolving difficulty conditions.

These results also suggest that the **FlexDiff** module, while crucial for adjusting difficulty, can lead to instability without the credit assignment mechanism provided by **CGRPA**. The **CGRPA** module, by evaluating each agent’s contribution through counterfactual reasoning, ensures that agents can adapt quickly and effectively to newly adjusted environments. This helps maintain stable convergence during difficulty transitions, preventing agents from falling into suboptimal strategies and allowing them to recover faster after performance drops.

2) *Performance Impact of Counterfactual Relative Advantage Weight λ* : We further examine how the choice of key hyperparameter λ , i.e., the optimization objective weight of our **CGRPA** algorithm affect the overall performance of CL-MARL. To evaluate the learning quality and performance in a more quantitative manner, we introduce a new performance metric known as the *normalized win-rate*¹².

As shown in Fig. 9, we independently ran 5 trials with maximum 200,000 steps on three maps at three different difficulty levels, and obtained the average normalized win-rate, corresponding to parts (a), (b), and (c). The dashed lines represent the normalized win-rate of original QMIX, serving as a baseline performance reference. From these results, we can observe that the choice of λ has a significant impact on performance. For instance, on the Hard maps $8m_vs_9m$ and $3s_vs_5z$, as well as all Super-hard maps, lower values of λ notably reduce the final win-rate. Moreover, when λ is set to -1, we observe a substantial negative impact on performance, even resulting in worse results than the original QMIX. This further empirically demonstrates that the counterfactual group-relative advantage can effectively enhance the credit assignment of the Q-value in the AC-based MARL algorithms, helping the agent continuously learn the optimal behavior policy from historical experiences.

3) *Investigation of Performance Bottlenecks in Some Super-Hard Scenarios*: To further examine the causes of CL-MARL’s suboptimal performance on certain *Super-Hard* maps, we applied the converged policy networks during real-time adversarial simulations. In most cases, the trained agents consistently outperformed the script-controlled opponents. This

¹²For each round of training with N evaluations, the normalized win-rate records the *top* - K highest values (with K set to 20 by default in our experiments) from the N win rates and computes their average as the final measurement. The purpose of this metric is to assess both the robustness of the learning process and the quality of the learning policy across different seed cases.

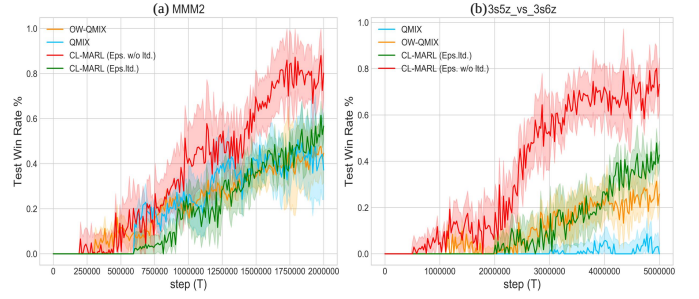


Fig. 10: Performance Comparison between CL-MARL and its variants with different episode limits: CL-MARL (Eps.Ltd.) is the original setup, while CL-MARL (Eps. w/o Ltd.) represents the results after increasing the maximum episode length.

observation suggests that the suboptimal performance might be due to an insufficient *Maximum Episode Length* instead of inadequate policy learning. Specifically, we hypothesized that the short step sizes, which were based on unreasonable prior assumptions during the win rate calculation for some Super-Hard maps, could have affected the computation of win rate.

Based on this insight, we explored the impact of the hyperparameter ‘*episode limit*’ on the training process. We conducted comparative experiments on the MMM2 and $3s5z_vs_3s6z$ maps, adjusting the *episode limit* hyperparameter from default 180 and 170 to 250 and 300. The version without the episode length limitation was labeled as CL-MARL (Eps. w/o Ltd.), while the original method was referred to as CL-MARL (Eps.Ltd.). We repeated the experiments five times independently and recorded the average performance. As shown in Fig. 10, we observed that CL-MARL (Eps. w/o Ltd.) significantly outperformed the default CL-MARL (Eps.Ltd.) on both maps, in terms of both convergence speed and final win rate. Moreover, it also outperformed other baselines such as OW-QMIX and QMIX.

In addition, after expanding the maximum episode length, we simulated adversarial interactions with the policy networks trained on the MMM2 and $3s5z_vs_3s6z$ maps, as shown in Fig. 11-12. It was evident that, under the extended simulation steps, the units controlled by our CL-MARL method maintained a consistent advantage over the opponents. However, under the original default simulation steps (i.e., 180 and 170), the agents did not fully defeat the opponents. As the simulation continued, the agents eventually triumphed at the moments corresponding to simulation step lengths of 220 and 200, defeating all opponent units. Therefore, we can conclude that the lower win rate of our CL-MARL algorithm was not due to its inability to outperform the opponents, but rather due to the initial step length limitation. This further validates the superiority and effectiveness of our proposed method.

VI. CONCLUSION AND FUTURE WORK

In this work, we identified and addressed such a critical challenge of “*environmental meta-stationarity*” in MARL, where fixed-difficulty training conditions limited policy generalization and trapped agents in suboptimal solutions. Specifically, we introduced a novel framework that integrates dy-

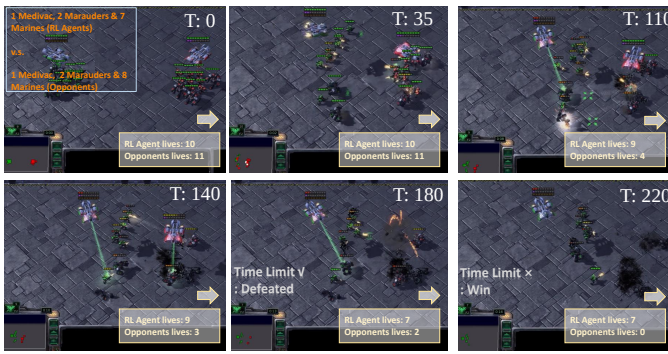


Fig. 11: Example of the Remaining Survival Status of Units in Real-Time Adversarial Simulations on the MMM2 Map, Based on the Number of Steps (T).



Fig. 12: Example of the Remaining Survival Status of Units in Real-Time Adversarial Simulations on the 3s5z_vs_3s6z Map, Based on the Number of Steps (T).

dynamic CL with MARL to overcome the limitations of environmental meta-stationarity. Specifically, our method combined two key innovations: (1) **FlexDiff**, an self-adaptive difficulty scheduler that dynamically modulates task complexity based on real-time agent performance, and (2) **CGRPA**, a novel credit assignment mechanism that merges *group relative policy optimization* with *counterfactual action advantage estimation* to maintain learning stability during curriculum transitions. Extensive experiments on the **SMAC** benchmark demonstrated significant improvements in both final performance and training efficiency compared to static-difficulty MARL baselines. Remarkably, our proposed approach demonstrated straightforward *plug-and-play compatibility* with baseline MARL algorithms, requiring only minimal architectural modifications while achieving superior task performance at the same training steps without compromising convergence speed.

In future work, we will focus on the following directions: First, we will explore more intelligent environment difficulty scheduling methods, including novel environment difficulty setting and grading approaches, to enhance the application of CL in MARL. Second, while our method has demonstrated strong performance in cooperative adversarial scenarios, we will extend its application to other MARL scenarios to thoroughly evaluate our method's generalizability. Third, we will investigate heterogeneous multi-agent systems with diverse agent roles and capabilities to advance adaptive MARL frame-

works. Meanwhile, we will also attempt to scale to large-scale systems and enhancing sample efficiency for real-world deployment, thereby further validating the effectiveness of CL in MARL applications.

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